

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

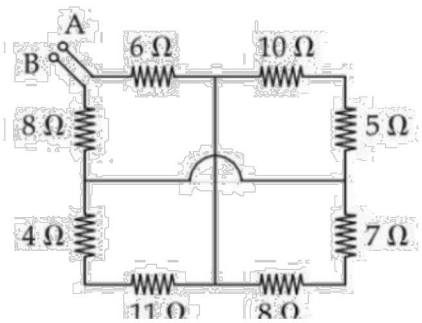
Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-17** **Date: 11.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The equivalent resistance between A and B is:



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A) 19Ω B) 25Ω C) 27Ω D) 18Ω

2. A sample of 1 mole gas at temperature T is adiabatically expanded to double its volume. If adiabatic constant for the gas is $\gamma = \frac{3}{2}$, then the work done by the gas in the process is :

A) $RT[2+\sqrt{2}]$ B) $\frac{T}{R}[2+\sqrt{2}]$ C) $\frac{R}{T}[2-\sqrt{2}]$ D) $RT[2-\sqrt{2}]$

3. Given below are two statements:

Statements (I): When an object is placed at the center of curvature of a concave lens, image is formed at the center of curvature of the lens on the other side.

Statement (II): Concave lens always forms a virtual and erect image.

In the light of the above statements, choose the correct Answer from the options given below:

A) Statement I is false but statement II is true

B) Both statement I and statement II are true

C) statement I is true but statement II is false

D) Both statement I and statement II are false

4. The dimensional formula of latent heat is:

A) $[M^0L^2T^{-2}]$ B) $[MLT^{-2}]$ C) $[ML^2T^{-2}]$ D) $[M^0LT^{-2}]$

5. Given below are two statements:

Statement (I): When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.

Statement (II): Ampere's circuital law does not depend on Biot-Savart's Law.

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In the light of the above statements, choose the correct answer from the options given below:

A) statement I is true but statement II is false

B) Statement I is false but Statement II is true

C) Both statement I and statement II are true

D) Both statement I and statement II are false

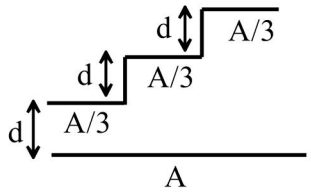
6. A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is :

A) $\left(\frac{\sigma+1}{\sigma-1}\right)^{\frac{1}{3}}$ B) $\left(\frac{\sigma-2}{\sigma+2}\right)^{\frac{1}{3}}$ C) $\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$ D) $\left(\frac{\sigma-1}{\sigma}\right)^{\frac{1}{3}}$

7. One main scale division of a Vernier caliper is equal to m units. If n^{th} division of main scale coincides with $(n+1)^{th}$ division of Vernier scale , the least count of the Vernier caliper is :

A) $\frac{n}{(n+1)}$ B) $\frac{1}{(n+1)}$ C) $\frac{m}{(n+1)}$ D) $\frac{m}{n(n+1)}$

8. A capacitor is made of a plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d, the capacitance of the arrangement is :



A) $\frac{13\varepsilon_0 A}{17d}$ B) $\frac{11\varepsilon_0 A}{18d}$ C) $\frac{18\varepsilon_0 A}{11d}$ D) $\frac{11\varepsilon_0 A}{20d}$

9. An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of 318.5 km. From earth's surface to the orbit,

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the change in total mechanical energy of the ball is $X\frac{GM_em}{21R_e}$. The value of x is (take $R_e = 6370$ km):

A) 11 B) 9 C) 10 D) 12

10. A particle of mass m moves on a straight line with its velocity increasing with distance according to the equation $v = \alpha\sqrt{x}$, where α is a constant. The total work done by all the forces applied on the particle during its displacement from $x = 0$ to $x = d$ will be:

A) $\frac{md}{2\alpha^2}$ B) $\frac{m\alpha^2 d}{2}$ C) $2m\alpha^2 d$ D) $\frac{m^2}{2\alpha^2 d}$

11. A galvanometer has a coil of resistance 200Ω with a full scale deflection at $20\mu A$. The value of resistance to be added to use it as an ammeter of range $(0-20)$ mA is:

A) 0.20Ω B) 0.10Ω C) 0.50Ω D) 0.40Ω

12. A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is $\frac{g}{8}$, then the ratio of the masses $\frac{m_2}{m_1}$ is :

A) 8 : 1 B) 9 : 7 C) 4 : 3 D) 5 : 3

13. A proton, an electron and an alpha particle have the same energies. Their de-Broglie wave lengths will be compared as:

(A) $\lambda_p > \lambda_e > \lambda_\alpha$ B) $\lambda_e > \lambda_\alpha > \lambda_p$ C) $\lambda_p > \lambda_e > \lambda_\alpha$ D) $\lambda_\alpha < \lambda_p < \lambda_e$

14. A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb:

A) Decreases B) increases

C) Remains same D) becomes Zero

APEX JEE-MAIN 2025

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15. A plane EM wave is propagating along X direction. It has a wavelength of 4 mm. If electric field is in y direction with the maximum magnitude of $60Vm^{-1}$, the equation for magnetic field is :

A) $B_z = 60\sin\left[\frac{\pi}{2}(x-3\times10^8 t)\right]\hat{k}T$

B) $B_x = 60\sin\left[\frac{\pi}{2}(x-3\times10^8 t)\right]\hat{i}T$

C) $B_z = 2\times10^{-7}\sin\left[\frac{\pi}{2}(x-3\times10^8 t)\right]\hat{k}T$

D) $B_z = 2\times10^{-7}\sin\left[\frac{\pi}{2}\times10^3(x-3\times10^8 t)\right]\hat{k}T$

16. If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a}+\vec{b}|=\sqrt{2}|\vec{a}-\vec{b}|$ for $|\vec{a}|=n|\vec{b}|$. The integer value of n is ____

A) 4 B) 7 C) 9 D) 3

17. A light emitting diode (LED) is fabricated using GaAs semi conducting material whose band gap is 1.42 eV. The wavelength of light emitted from the LED is:

A) 1243nm B) 1400 nm C) 875 nm D) 500 nm

18. A particle moving in a straight line covers half the distance with speed 6m/s . The other half is covered in two equal time intervals with speeds 9 m/s and 15 m/s respectively. The average speed of the particle during the motion is :

A)8 m/s B) 8.8m/s C) 10m/s D) 9.2m/s

19. The volume of an ideal gas $(\gamma = 1.5)$ is changed adiabatically from 5 liters to 4 liters. The ratio of initial pressure to final pressure is:

A) $\frac{16}{25}$ B) $\frac{8}{5\sqrt{5}}$ C) $\frac{4}{5}$ D) $\frac{2}{\sqrt{5}}$

APEX JEE-MAIN 2025

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20. The energy equivalent of 1g of substance is:

A) $5.6eV$ B) $5.6\times10^{26}MeV$ C) 11.2×10^{24}MeV D) $5.6\times10^{12}MeV$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

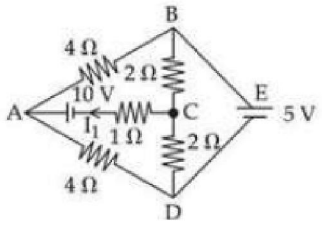
21. The position, velocity and acceleration of a particle executing simple harmonic motion are found to have magnitudes of 4m, 2ms^{-1} and 16 ms^{-2} at a certain instant. The amplitude of the motion is $\sqrt{x}\ m$ where x is ____

22. In a young's double slit experiment , the intensity at a point is $\left(\frac{1}{4}\right)^{th}$ of the maximum intensity, the minimum distance of the point from the central maximum is _____ μm. (Given $\lambda=600nm, d=1.0mm, D=1.0m$)

23. Two persons pull a wire towards themselves. Each person exerts a force of 200N on the wire. Young's modulus of the material of wire is $1\times10^{11}Nm^{-2}$ .Original length of the wire is 2 m and the area of cross section is $2cm^2$. The wire will extended in length by _____ μm.

24. At the center of a half ring of radius R=10 cm and linear charge density 4n Cm^{-1} , the potential is $x\pi V$. The value of x is ___

25. The current flowing through the 1Ω resistor is $\frac{n}{10}A$. The value of n is _____



APEX JEE-MAIN 2025

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CHEMISTRY **MAX.MARKS: 100**

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(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The electronic configuration of $Cu(II)$ is $3d^9$ whereas that of $Cu(I)$ is $3d^{10}$. Which of the following is correct?

A) $Cu(I)$ and $Cu(II)$ are equally stable

B) $Cu(II)$ is less stable

C) $Cu(II)$ is more stable

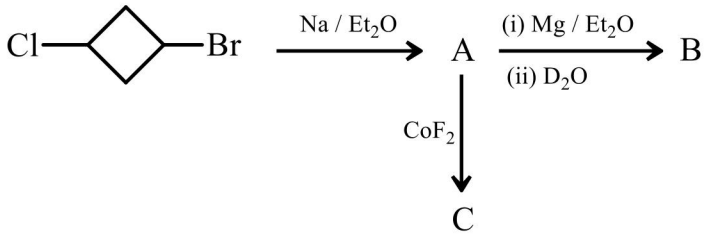
D) stability of $Cu(I)$ and $Cu(II)$ depends on nature of copper salts

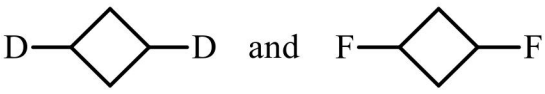
27. The F^- ions make the enamel on teeth much harder by converting hydroxyapatite (the enamel on the surface of teeth) into much harder fluoropatite having the formula.

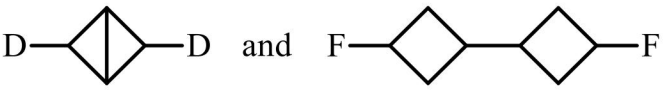
A) $[Ca_3(PO_4)_2.3Ca(OH)_2]$ B) $[3(Ca_3(PO_4)_2).CaF_2]$

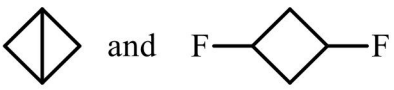
C) $[3(Ca_3(PO_4)_2).Ca(OH)_2]$ D) $[Ca_3(PO_4)_2.3CaF_2]$

28. In the following sequence of reaction, the major products B and C respectively are:



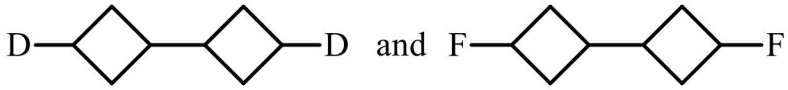
A) 

B) 

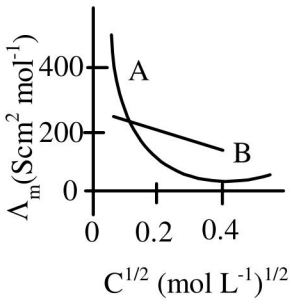
C) 

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

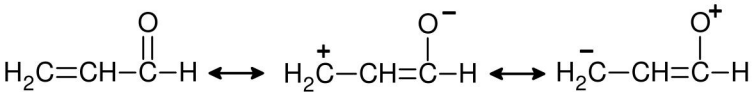
D) 

29. The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are:



	A	**B**
A)	Weak electrolyte	Weak electrolyte
B)	Strong Electrolyte	Strong electrolyte
C)	Strong electrolyte	Weak electrolyte
D)	Weak electrolyte	Strong electrolyte

30. Relative stability of the contributing structure is:



(I) (II) (III)

A) $(I)>(III)>(II)$ B) $(II)>(I)>(III)$

C) $(I)>(II)>(III)$ D) $(III)>(II)>(I)$

31. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion(A): Both rhombic and monoclinic Sulphur exist as S_8 while oxygen exists as O_2.

Reason (R): Oxygen forms $p\pi - p\pi$ multiple bonds with itself and other elements having small size and high electronegativity like C, N, which is not possible for Sulphur.

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

In the lihgt of the above statements, choose the most appropriate answer from the options given below.

A) Both (A) and (R) are correct and R is the correct explanation of (A).

B) Both (A) and (R) are correct but R is not the correct explanation of (A).

C) (A) is correct but (R) is not correct.

D) (A) is not correct but (R) is correct.

32. Given below are two statements:

Statement (I): The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.

Statement (II): $p\pi - p\pi$ bonds formation is more prevalent in second period elements over other periods.

In the light of the above statements choose the most appropriate answer from the options given below:

A) Statement I is correct but statement II is incorrect

B) Both statement I and statement II are correct

C) Both statement I and statement II are incorrect

D) Statement I is incorrect but statement II is correct

33. 0.05M $CuSO_4$ when treated with 0.01 M $K_2Cr_2O_7$ given green color solution of $Cu_2Cr_2O_7$ the two solutions are separated as shown below;

[SPM: semi permeable Membrane]

$K_2Cr_2O_7$		$CuSO_4$
Side X	SPM	Side Y

Due to osmosis:

A) Molarity of $CuSO_4$ solution is lowered.

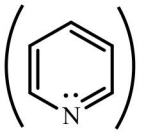
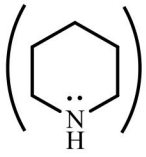
B) Green colour formation observed on side Y.

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

C) Molarity of $K_2Cr_2O_7$ solution is lowered

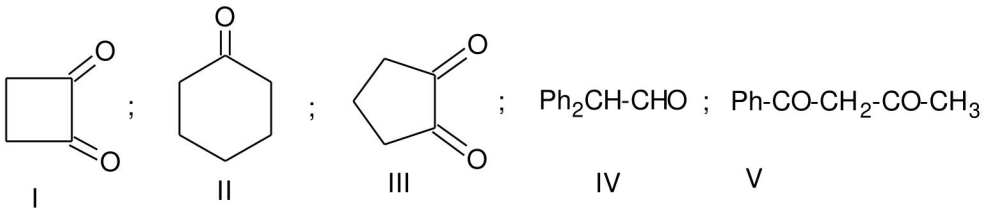
D) Green colour formation observed on side X.

34. Correct order of basic strength of pyrrole  ,Pyridine  and piperidine  is

A) pyrrole>pyridine>piperidine B) piperidine>pyridine>pyrrole

C) pyridine>piperidine>pyrrole D) pyrrole>piperidine>pyridine

35. Which of the following is not correct about the % enol of the following compounds?



A) V > IV > II B) I > II > IV C) III > I > II D) III > V > IV

36. Given below are two statements:

Statement-1 : Beryllium salts are acidic when dissolved in pure water

Statement-2 : Be–O bond is very strong thus it weakens the O–H bond in hydrated ion

A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

C) Statement-1 is True, Statement-2 is False

D) Statement-1 is False, Statement-2 is True

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

37. Given below are two statements: one is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): S_N2 reaction of $C_6H_5CH_2Br$ occurs more readily than the S_N2 reaction of CH_3CH_2Br

Reason (R): The partially bonded hybridized p-orbital that develops in the trigonal bipyramidal transition state is stabilized by conjugation with the phenyl ring

In the light of the above statements, choose the most appropriate answer form the options given below:

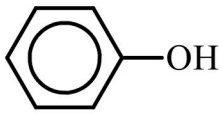
A) (A) is not correct but (R) is correct

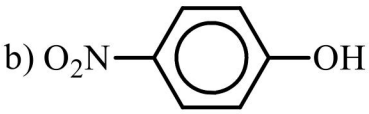
B) (A) Is correct but (R) is not correct

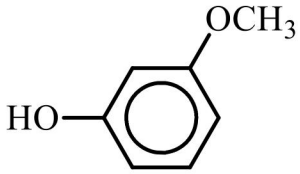
C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

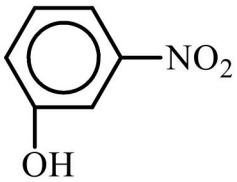
D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

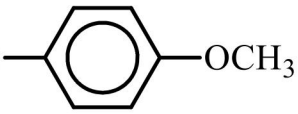
38. For the given compounds, the correct order of increasing pk_a value:

a) 

b) 

c) 

d) 

e) 

choose the correct answer from the options given below :

A) $(e)>(d)<(b)<(a)<(c)$

B) $(e)<(d)<(c)<(b)<(a)$

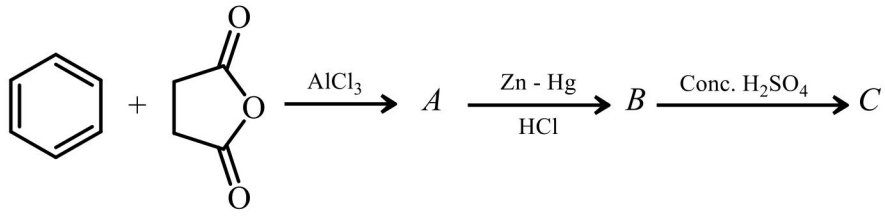
C) $(d)<(e)<(c)<(b)<(a)$

D) $(b)<(d)<(c)<(a)<(e)$

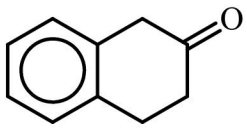
APEX JEE-MAIN 2025

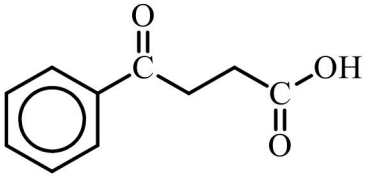
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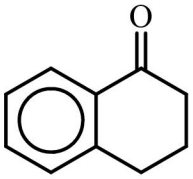
39.

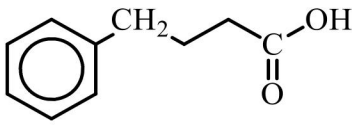


What is the structure of C?

A) 

B) 

C) 

D) 

40. Identify the incorrect statement regarding primary standard of titrimetric analysis:

a) If should be purely available in dry form.

b) It should not undergo chemical change in air.

c) It should be hygroscopic and should react with another chemical instantaneously and stoichiometrically

d) It should be readily soluble in water

e) $KMnO_4$ & $NaOH$ can be used as primary standard

Choose the correct answer from the options given below:

A) (b) and (e) only

B) (c) and (d) only

C) (c) and (e) only

D) (a) and (b) only

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

41. Given below are two statements: One is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A) : The total number of geometrical isomers shown by $[Co(en)_2Cl_2]^+$ complex ion is three.

Reason (R) : $[Co(en)_2Cl_2]^+$ complex ion has an octahedral geometry.

In the light of the above statements, choose the most appropriate answer from the options given below:

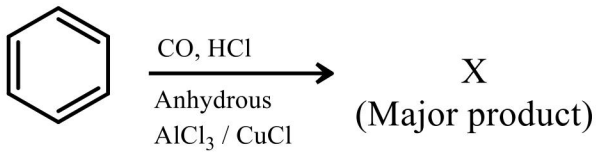
A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

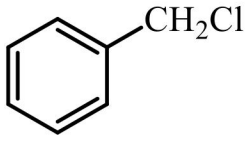
B) (A) is not correct but (R) is correct

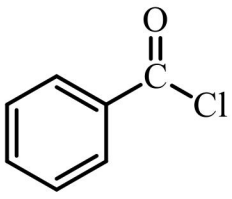
C) (A) is correct but (R) is not correct

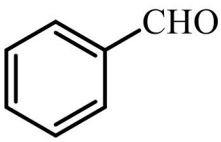
D) Both (A) and (R) are correct and (R) is the correct explanation of (A).

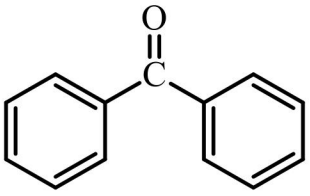
42. Identify major product ''X'' formed in the following reaction:



A) 

B) 

C) 

D) 

43. In which of the follwing pairs the central atoms exhibit sp^2 hybridization?

A) BF_3 and NO_2^-

B) BF_3 and H_2O

C) NH_2^- and H_2O

D) NH_2^- and BF_3

APEX JEE-MAIN 2025

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44. Methods used for purification of organic compounds are based on:

A) Nature of compound and presence of impurity

B) Nature of compound only

C) Neither on nature of compound nor on the impurity present

D) Presence of impurity only

45. Compare the energies of following sets of quantum numbers for multielectron system.

a) $n=4, 1=1$ b) $n=4, l=2$ c) $n=3, l=1$ d) $n=3, l=2$ e) $n=4, 1=0$

Choose the correct answer from the options given below:

A) $(b)>(a)>(d)>(c)>(e)$ B) $(e)>(c)<(d)<(a)<(b)$

C) $(e)>(c)>(a)>(d)>(b)$ D) $(c)>(e)>(d)>(a)>(b)$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Number of ambidentate ligands among the following is _____

$NO_2^-, SCN^-, C_2O_4^{2-}, NH_3, CN^-, SO_4^{2-}, H_2O$

47. The standard reduction potentials at 298 k for the following half cells are given below:

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O, E^0 = 1.33V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe, E^0 = -0.04V$

$Ni^{2+}(aq) + 2e^- \rightarrow Ni, E^0 = -0.25V$

$Ag^+(aq) + e^- \rightarrow Ag, E^0 = 0.80V$

$Au^{3+}(aq) + 3e^- \rightarrow Au, E^0 = 1.40V$

Consider the given electro chemical reactions, the number of metal(s) which will be oxidized by $Cr_2O_7^{2-}$, in aqueous solution is _____

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

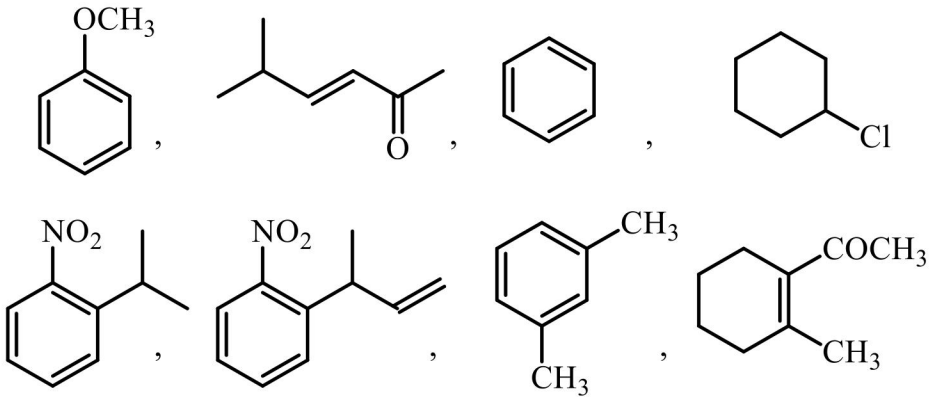
48. Total number of essentials amino acid among the given list of amino acids is ___

Arginine, phenylalanine, Aspartic acid, Cysteine, Histidine, Valine, Proline

49. Number of colorless lanthanide ions among the following is _____

$Eu^{3+}, Lu^{3+}, Nd^{3+}, La^{3+}, Sm^{3+}$

50. How many compounds among the following compounds show inductive, mesomeric as well as hyper conjugation effects?



MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $\overline{OA} = 2\bar{a}, \overline{OB} = 6\bar{a} + 5\bar{b}$ and $\overline{OC} = 3\bar{b}$ where O is the origin . If the Area of the parallelogram with adjacent sides $\overline{OA}$ and $\overline{OC}$ is 15 sq. units , then the area (in sq. units) of the quadrilateral OABC is equal to

A) 40 B) 38 C) 32 D)35

52. The solution of the differential equation $(x^2 + y^2)dx - 5xydy = 0, y(1) = 0$ is

A) $|x^2 - 4y^2|^6 = x$ B) $|x^2 - 2y^2|^5 = x^2$ C) $|x^2 - 2y^2|^6 = x$ D) $|x^2 - 4y^2|^5 = x^2$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

53. The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+....+x^{54}(1+x)^{46}$ is ${}^{99}C_p-{}^{46}C_q$ then a possible value of $p+q$ is

A) 68 B) 61 C) 55 D)83

54. Let the line L intersect the lines $x-2=-y=z-1, 2(x+1)=2(y-1)=z+1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$ then which of the following point lies on L?

A) $\left(\frac{-1}{3},1,-1\right)$ B) $\left(\frac{-1}{3},-1,-1\right)$ C) $\left(\frac{-1}{3},-1,1\right)$ D) $\left(\frac{-1}{3},1,1\right)$

55. Let three vectors $\bar{a}=\alpha\hat{i}+4\hat{j}+2\hat{k}, \bar{b}=5\hat{i}+3\hat{j}+4\hat{k}, \bar{c}=x\hat{i}+y\hat{j}+z\hat{k}$ form a triangle such that $\bar{c}=\vec{a}-\vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to

A)16 B) 14 C) 12 D) 10

56. A variable line L passes through the point (3,5) and intersects the positive co-ordinate axes at the points A and B. The minimum area of the triangle OAB, where O is the origin is

A)25 B) 35 C) 30 D) 40

57. The parabola $y^2=4x$ divides the area of the circle $x^2+y^2=5$ in two parts. The area of the smaller part is equal to

A) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ B) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

C) $\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ D) $\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

58. If the sum of the series $\frac{1}{1.(1+d)}+\frac{1}{(1+d)(1+2d)}+....+\frac{1}{(1+9d)(1+10d)}$ is equal to 5, then 50 d is equal to

A) 10 B) 20 C) 5 D)15

APEX JEE-MAIN 2025

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59. The shortest distance between the lines $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$

A) $\frac{185}{\sqrt{563}}$ B) $\frac{178}{\sqrt{563}}$ C) $\frac{187}{\sqrt{563}}$ D) $\frac{179}{\sqrt{563}}$

60. Let $f(x)=x^2+9, g(x)=\frac{x}{x-9}$ and $a=fog(10), b=gof(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a}+\frac{y^2}{b}=1$ then $8e^2+l^2$ is equal to

A) 16 B) 6 C) 12 D) 8

61. Let $f(x)=ax^3+bx^2+cx+41$ be such that $f(1)=40, f'(1)=2, f''(1)=4$ then $a^2+b^2+c^2$ is equal to

A) 51 B) 62 C) 73 D) 54

62. Let a circle passing through (2,0) have is center at the point (h,k). Let (x_c, y_c) be the point of intersection of the lines $3x+5y=1$ and $(2+c)x+5c^2y=1$ If $h=\lim_{c\to 1} x_c$ and $k=\lim_{c\to 1} y_c$ then the equation of the circle is

A) $25x^2+25y^2-2x+2y-60=0$ B) $5x^2+5y^2-4x-2y-12=0$

C) $5x^2+5y^2-4x+2y-12=0$ D) $25x^2+25y^2-20x+2y-60=0$

63. The solution curve, of the differential equation $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$ passing through the point (0, 1) is a conic , whose vertex lies on the line

A) $2x+3y=9$ B) $2x+3y=-9$ C) $2x+3y=-6$ D) $2x+3y=6$

APEX JEE-MAIN 2025

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64. A ray of light coming from the point $P(1,2)$ gets reflected from the point Q on the x-axis and then passes through the point $R(4,3)$. If the point $S(h,k)$ is such that PQRS is a parallelogram, then hk^2 is equal to

A)60 B) 90 C) 80 D) 70

65. The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	20
No. of Students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x+5y$ is equal to

A) 46 B) 43 C) 47 D) 44

66. If the domain of the function $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $R-(\alpha,\beta)$ then $12\alpha\beta$ is equal to

A) 24 B) 36 C) 40 D) 32

67. Let $\lambda,\mu\in R$ if the system of equations $3x+5y+\lambda z=3, 7x+11y-9z=2, 97x+155y-189z=\mu$ has infinitely many solutions, then $\mu+2\lambda$ is equal to

A) 22 B) 25 C) 24 D) 27

68. Let $\int\frac{2-\tan x}{3+\tan x}dx=\frac{1}{2}(\alpha x+\log_e|\beta\sin x+\gamma\cos x|)+C$, where C is the constant of integration then $\alpha+\frac{\gamma}{\beta}$ is equal to

A) 3 B) 4 C) 7 D) 1

69. Let α,β be the roots of the equation $x^2+2\sqrt{2}x-1=0$. The quadratic equation, whose roots are $\alpha^4+\beta^4$ and $\frac{1}{10}\left(\alpha^6+\beta^6\right)$ is

A) $x^2-180x+9506=0$ B) $x^2-190x+9466=0$

APEX JEE-MAIN 2025

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C) $x^2-195x+9506=0$ D) $x^2-195x+9466=0$

70. Let $|\cos\theta\cos(60-\theta)\cos(60+\theta)|\le\frac{1}{8},\theta\in[0,2\pi]$. Then , the sum of all $\theta\in[0,2\pi]$, where $\cos3\theta$ attains its maximum value is

A) 6π B) 9π C) 18π D) 15π

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and −1 if not correct.

71. Let $A=\{2,3,6,7\}B=\{4,5,6,8\}$. Let R be a relation defined on $A\times B$ by $(a_1,b_1)R(a_2,b_2)$ if and only if $a_1+a_2=b_1+b_2$, then the number of elements in R is ____.

72. Let $f:(0,\pi)\to R$, be a function given by

$$f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8 & x=\frac{\pi}{2}\\ (1+|\cot x|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$$

where $a,b\in Z$, if f is continuous at $\frac{\pi}{2}$, then a^2+b^2 is equal to ___

73. The sum of the square of the modulus of the elements in the $\{z=a+ib:\ a,b\in Z,\ z\in C,\ |z-1|\le1,\ |z-5|\le|z-5i|\}$ is _____

74. Let A be a non-singular matrix of order 3, if $\det(3adj(2adj((\det A)A)))=3^{-13}.2^{-10}$ and $\det(3adj(2A))=2^m.3^n$ then $|3m+2n|$ is equal to _____

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75. Let a, b and c denote the outcome of three independent rolls of a fair tetrahedral die, whose four faces are marked 1, 2, 3, 4. If the probability that $ax^2+bx+c=0$ has all real roots is $\frac{m}{n}, \gcd(m,n)=1$ then $m+n$ is equal to _____

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